INCH-POUND

MIL-DTL-60286C
12 September 2000
SUPERSEDING
MIL-I-60286B
21 September 1992

DETAIL SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, M53 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M53 Cartridge Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-2550	Ammunition, General Specification For
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

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MIL-DTL-48412	Initiator, Cartridge Actuated, Delay, M53 Remanufactured Metal Parts Assembly
MIL-DTL-60519	Initiator, Cartridge Actuated, M53 Metal Parts Assembly

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of Department of Defense specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8597230	Cartridge, Impulse, M91 Assembly
10520619	Initiator, Cartridge Actuated, M53 Assembly

(Application for copies should be addressed to Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

PUBLICATIONS

AMERICAN SOCIETY FOR QUALITY CONTROL

ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes
ASQC A8402	Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Examination

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Components.

3.2.1 Primary components. For the purpose of this specification, the M91 impulse cartridge (Drawing 8597230) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

3.2.2 Component parts. All component parts shall meet the applicable requirements of MIL-DTL-48412 or MIL-DTL-60519 prior to assembly.

3.3 Breakaway torque. The chamber and cap of each assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.3.1).

3.4 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction (see 4.3.2).

3.5 Radiographic examination. Each initiator assembly shall be radiographically examined for defects (see 4.3.3).

3.6 Functional.

3.6.1 Ballistics. Initiators loaded with cartridges from one accepted lot and temperature conditioned shall perform within the pressure values and delay times (see 6.4.1) specified in table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, assembly number MS28741-4, and a pressure chamber with a volume of 1 cubic inch.

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3.6.1.1 Initiation. The initiator shall be fired by a source delivering an output pressure of 700 +25, -0 pounds per square inch gage (psig) at 70 ± 5 °F.

3.6.2 Misfires. There shall be no misfires.

3.6.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.4.2).

TABLE I. Pressure values.

Temperature (°F)	Pressure (psig)		Ignition delay milliseconds, max
	Minimum	Maximum	
-60 to -70	500	1425	65
65 to 75	600	1550	65
195 to 205	700	1700	65

3.7 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

4.1.2 Quality assurance terms and definitions. Reference shall be made to ASQC A8402 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and ASTM E1742.

4.2 Conformance inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission or product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with ASQC Z1.4.

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4.2.1.2 Size of lot. The production lot size shall be as specified in the contract (see 6.2).

4.2.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.2.2.1 Cartridge, Drawing 8597230, covering a detail of Drawing 10520619.

Characteristics	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing 10520619, less all drawings except Drawings 10520473, 10520474, 10520622, 10520496, and MS29513-9.

Characteristics	Defect	Method of Inspection
Critical		
1	Firing pin protrusion	Gage
2	Shear pin not engaged in firing pin	Gage
3	Striker hole in initiator cap obstructed	Visual

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Major		
101	O-ring not lubricated	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
None defined		

4.2.2.3 Assembly, Drawing 10520619, less all drawings except Drawings 10520407, 10520414, and 8597230.

Characteristics	Defect	Method of Inspection
Critical		
1	Cartridge not properly seated	Visual
2	Filter missing or insecure	Visual
3	Nylon patch missing or improperly applied	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.2.2.4 Assembly, Drawing 10520619.

Characteristics	Defect	Method of Inspection
Critical		
1	Cartridge face not seated against base of firing pin housing	Visual/X-ray
2	Any part missing	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, incorrect, or illegible	Visual

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Major		
101	Set screw missing or insecure	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
201	Thread sealing compound missing from setscrews	Visual
202	Protective cap or plug missing	Visual
203	Protective finish inadequate or defective	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with 3.3 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with 3.4 shall be cause for rejection of the initiator.

4.2.3.3 Functional tests. The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.6. The units shall be temperature conditioned as provided in table II and 4.3.3.2. Failure of any unit to comply with 3.6 shall be cause for rejection of the lot.

TABLE II. Conditioning temperature.

Temperature (°F)	Sample Size
-60 to -70	7
65 to 75	7
195 to 205	7

4.2.3.3.2 Retest. There shall be no retests.

4.2.3.3.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

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4.3.1 Breakaway torque. The torque wrench set as specified in 3.3 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in a suitable vise, and the torque shall be applied to the cap.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in the north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.3.3 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with ASTM E1742, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The initiators shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Pin, Firing, P/N 10520474
b. O-ring, MS29513-9
c. Filter, P/N 10520414
d. O-ring, P/N 10520496
e. Pin, Shear, P/N 10520622
f. Shear Pin Setscrew
g. Setscrew, Initiator Cap
h. Cartridge, Impulse, M91, P/N 8597230

4.3.4 Functional.

4.3.4.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired by a source delivering an output of 700 +25, -0 psig at the end of 15 feet of aircraft hose, MS28741-4, at 70 ± 5°F. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.4.2. Required pressure readings shall be taken as specified in 3.6.1.

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4.3.4.2 Temperature conditioning. Each assembled initiator shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -65°F, all condensation shall be removed from the exterior surfaces before returning the initiator to the conditioning chamber.

4.4 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. As the Aircrew Emergency Escape Sytems are designed only for military aircraft, there is no commercial application for the M53 Cartridge Actuated Initiator Assembly

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2 and 2.3)
c. Production test activity and lot size (see 4.2.1)
d. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.3)
e. Level of packing and marking (see 5.1and 6.5)
f. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 Definitions.

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6.4.1 Ignition delay. Ignition delay time is the time measured between the firing pin striking the primer or ignition device and the beginning of a continuous thrust rise on a thrust-time record.

6.4.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5. Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.5.1 Packing and marking. Packing and marking should be Level A or B, as specified by the contracting activity (see 6.2).

6.5.1.1 Level A. Level A should be in accordance with SPI F00-138-3770.

6.5.1.2 Level B. Level B should be in accordance with SPI F00-138-3770.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity Navy - OS
(Project 1377-0137)

Review activity:
Air Force - 70

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-60286C	**2. DOCUMENT DATE (YYYYMMDD)** 20000912

3. DOCUMENT TITLE **INITIATOR, CARTRIDGE ACTUATED, M53 ASSEMBLY**

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION, NAVAL SURFACE WARFARE CENTER	b. TELEPHONE *Include Area Code)* (1) Commercial (301) 744-4700 (2) AUTOVON 354-4700
c. ADDRESS *(Include Zip Code)* STANDARDIZATION TEAM (CODE 840M) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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